# A RELATIVE DUALIZING COMPLEX IS UNIQUE UP TO A UNIQUE ISOMORPHISM

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0E2W (sources/stacks-duality/pinned-original-source.tex line 7334).

*Allowed standard prerequisites (editorial).*

- The affine algebraic uniqueness of the relative dualizing complex. Same chapter, Lemma labelled lemma-relative-dualizing-complex-algebra, cited at proof lines 7341-7344, and Dualizing Complexes, Lemma labelled dualizing-lemma-uniqueness-relative-dualizing, cited at proof lines 7344-7346. Provides the isomorphisms on affine opens.
- The automorphism group and negative Ext vanishing for a relative dualizing complex. Same chapter, Lemma labelled lemma-relative-dualizing-RHom, cited at proof lines 7358-7360 and 7366-7368. Shows there is at most one normalised isomorphism and gives the vanishing needed to glue.
- Glueing of morphisms from the vanishing of negative Ext groups. Cohomology, Lemma labelled cohomology-lemma-vanishing-and-glueing, cited at proof lines 7368-7372. Produces the global morphism agreeing with the local ones.
- The generator property of the comparison map defining a relative dualizing complex. The invertibility of the Delta_*O_X-module generated by eta and alpha(xi), used at proof lines 7350-7356. Allows alpha to be normalised by an invertible section.

*Source reference.* Original lemma 0E2U: sources/stacks-duality/pinned-original-source.tex, source lines 7271–7271. Same-chapter pointer into the author environment lemma at source line 7271; invoked at proof line 7344 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

---

*Date*: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0E2D: sources/stacks-chow/tags.txt, source lines 17603–17603. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7346 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0E2V: sources/stacks-duality/pinned-original-source.tex, source lines 7318–7318. Same-chapter pointer into the author environment lemma at source line 7318; invoked at proof line 7360 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0D66: sources/stacks-chow/tags.txt, source lines 16511–16511. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7371 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

**Lemma 0.1.** *Let $X \to S$ be a morphism of schemes which is flat and locally of finite presentation. If $(K, \xi)$ and $(L, \eta)$ are two relative dualizing complexes on $X/S$, then there is a unique isomorphism $K \to L$ sending $\xi$ to $\eta$.*

*Proof.* Let $U \subset X$ be an affine open mapping into an affine open of $S$. Then there is an isomorphism $K|_U \to L|_U$ by Lemma 0E2U and Dualizing Complexes, Lemma 0E2D. The reader can reuse the argument of that lemma in the schemes case to obtain a proof in this case. We will instead use a glueing argument.

Suppose we have an isomorphism $\alpha : K \to L$. Then $\alpha(\xi) = u\eta$ for some invertible section $u \in H^0(W, \Delta_*\mathcal{O}_X) = H^0(X, \mathcal{O}_X)$. (Because both $\eta$ and $\alpha(\xi)$ are generators of an invertible $\Delta_*\mathcal{O}_X$-module by assumption.) Hence after replacing $\alpha$ by $u^{-1}\alpha$ we see that $\alpha(\xi) = \eta$. Since the automorphism group of $K$ is $H^0(X, \mathcal{O}_X^*)$ by Lemma 0E2V there is at most one such $\alpha$.

Let $\mathcal{B}$ be the collection of affine opens of $X$ which map into an affine open of $S$. For each $U \in \mathcal{B}$ we have a unique isomorphism $\alpha_U : K|_U \to L|_U$ mapping $\xi$ to $\eta$ by the discussion in the previous two paragraphs. Observe that $\mathrm{Ext}^i(K|_U, K|_U) = 0$ for $i < 0$ and any open $U$ of $X$ by Lemma 0E2V. By Cohomology, Lemma 0D66 applied to $\mathrm{id} : X \to X$ we get a unique morphism $\alpha : K \to L$ agreeing with $\alpha_U$ for all $U \in \mathcal{B}$. Then $\alpha$ sends $\xi$ to $\eta$ as this is true locally. □